Josselin Davion-Jouffre

[email] | [phone] | [LinkedIn] | [GitHub]

6-month gap-year internship in quantitative finance / AI, from May 2027

Education

Mines Paris – PSL 2025 – 2029
MSc-level engineering degree (Ingénieur Civil) *Paris, France*

- Admitted ranked 24th in the national Mines-Ponts entrance exam (PSI track)
- Applied mathematics: stochastic calculus, probability, optimisation
- Data: data science, signal processing
- Gap year May 2027 – Aug 2028 before final year

ETH Zürich – ML for Quantitative Finance Oct 2026 – Mar 2027
Auditing student & self-directed training, alongside internship *Zurich, Switzerland*

- Audited course: *Machine Learning in Finance and Insurance* (Prof. P. Cheridito), with its 4 projects incl. deep hedging
- 22-week personal programme: classical ML and boosting, financial time series (GARCH), options (Hull), purged cross-validation and backtesting (López de Prado), sequential deep learning
- Code on GitHub: [link to a public version of the repo]

Lycée Lakanal – Classes Préparatoires 2023 – 2025
PSI – intensive two-year programme in maths, physics & engineering science* *Sceaux, France*

Experience

ETH Zürich – Lab of Prof. Dirk Mohr Sep 2026 – Feb 2027
Research Intern, Machine-Learning-Assisted Mechanics *Zurich, Switzerland*

- Finite element simulations of plasticity and dynamic material behaviour (Abaqus)
- Split Hopkinson bar experiments and processing of experimental data (Python)
- Neural-network prediction of mechanical response (PyTorch, TensorFlow)

Colibri May 2026 – present
Co-founder

- Platform helping students declare private tutoring through the French home-services tax credit
- Built end-to-end with a co-founder: software, business-process design, sales, marketing
- 400 users, about €20k of tutoring processed per month, €1.3k monthly revenue

Independent Tutor Sep 2025 – present
Self-employed

- Around ten high-school and classes préparatoires students in maths, physics and engineering science

Projects

Mines Paris engineering project – CEMEF Nov 2025
Team project *Sophia Antipolis, France*

- Designed from scratch a new hybrid electrode glass-furnace geometry
- Finite element modelling (fluid dynamics, heat transfer) in MSC Patran / Nastran, post-processing in ParaView
- Temperature control with adaptive-gain PID controllers: 30 % energy saving over previous designs

[Quant / ML project] [20XX]
[Context: personal, course, hackathon]

- [What you did, method, quantified result, GitHub link]

Skills

Programming: Python (NumPy, PyTorch, TensorFlow), SQL, Git, LaTeX
Languages: French (native), English (C1)
Interests: [2–3 specific items]